

SUMMARY

Results-driven Data Engineer with 4+ years of experience designing and implementing scalable cloud-based data platforms, machine learning–driven analytics systems, and real-time data pipelines. Specialized in system design, technical solution implementation, data architecture design, data integration, data management, and ML-ready pipeline development. Proven expertise in AWS, Azure, Spark, Snowflake, Kafka, Airflow, dbt, and MLOps platforms. Adept at data interpretation, engineering testing, and cross-functional collaboration to deliver production-grade AI/ML and analytics solutions.

SKILLS

- **Programming & Querying:** Python, SQL, PySpark, Scala, PowerShell
- **Cloud Platforms:** AWS (S3, Glue, Lambda, Redshift, EC2), Azure (Data Factory, Databricks, Synapse), GCP (BigQuery, Dataflow, Pub/Sub)
- **Big Data Processing:** Apache Spark, Hadoop
- **Streaming & Messaging:** Apache Kafka, AWS Kinesis, Azure Event Hubs, Pub/Sub
- **Data Warehouses & Storage:** Snowflake, Redshift, BigQuery, PostgreSQL, MySQL
- **Data Lakes & Lakehouse:** Delta Lake, Parquet, Medallion Architecture (Bronze/Silver/Gold)
- **Orchestration & Transformation:** Apache Airflow, dbt, Prefect
- **MLOps & ML Platforms:** MLflow, Kubeflow, SageMaker, TensorFlow, PyTorch
- **Feature Engineering:** Offline & Online Feature Stores, Feature Versioning
- **Data Modeling:** Star Schema, Snowflake Schema, Dimensional Modeling
- **Data Reliability & Governance:** Data Quality, Data Lineage, Data Observability, RBAC, Audit Logging
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, GitHub Actions
- **BI & Analytics:** Power BI, Tableau
- **APIs:** REST, GraphQL

EXPERIENCE

Meta	Sep2024 – Present
Data Engineer (AI/ML)	
• Designed and implemented scalable ELT pipelines using Spark, dbt, Snowflake, and Airflow, supporting batch and real-time ML workloads with 99.99% SLA. • Built centralized offline and online feature stores using AWS Glue, S3, and Delta Lake, enabling reusable features across 50+ ML models. • Architected real-time streaming systems using Kafka, Kinesis, and Spark Structured Streaming to support low-latency analytics and inference. • Led system design and end-to-end technical solution implementation for ML-ready data platforms. • Implemented medallion lakehouse architecture (Bronze, Silver, Gold) to standardize data ingestion, transformation, and consumption. • Optimized data warehouse performance using partitioning, clustering, and cost-based tuning, reducing query latency by 40% and cost by 30%. • Developed and tested model-ready datasets integrated with Kubeflow pipelines and SageMaker endpoints. • Built automated data validation, engineering testing frameworks, and pipeline monitoring to improve reliability. • Used PowerShell scripts to automate deployment tasks, environment validation, and data operations workflows. • Implemented CI/CD workflows using GitHub Actions for automated testing and deployment.	
Tools: AWS, Spark, Snowflake, Kafka, Airflow, dbt, Python, PowerShell, MLflow, SageMaker, GitHub Actions	
Microsoft	Sep 2020–Jul 2023
Data Engineer	
• Migrated legacy on-prem data pipelines to Azure (Data Factory, Databricks, Synapse), improving delivery time by 50%. • Designed distributed Spark-based data processing systems for large-scale data integration and transformation. • Implemented real-time ingestion pipelines using Azure Event Hubs and Kafka for streaming dashboards. • Designed dimensional data models and analytics layers to support business intelligence and reporting. • Delivered executive dashboards using Power BI for business stakeholders. • Built ML training and deployment pipelines using MLflow and Azure ML. • Supported data interpretation, KPI development, and ad-hoc analytics requests. • Enforced enterprise data governance, access controls, and compliance requirements.	
Tools: Azure Data Factory, Databricks, Synapse, Kafka, Event Hubs, Python, Airflow, MLflow, Azure ML, Power BI	

PROJECTS

HIPAA-Compliant AI Lakehouse Modernization

Technologies: AWS, Snowflake, Glue, S3, SageMaker, MLflow, Delta Lake

- Re-architected a legacy on-prem data warehouse into a cloud-native lakehouse platform.
- Designed scalable ingestion, transformation, and feature engineering pipelines.
- Implemented medallion architecture with raw, curated, and analytics-ready layers.
- Enabled end-to-end ML lifecycle management using MLflow and SageMaker.
- Reduced model retraining time by 40%.
- Enforced HIPAA compliance using encryption, RBAC, and audit logging.

LLM-Powered Data Quality Assistant

Technologies: FastAPI, LangChain, Great Expectations, LLMs

- Built a GenAI-powered system to automate data validation, anomaly detection, and rule generation.
- Implemented data observability pipelines for monitoring freshness, volume, and schema drift.
- Exposed real-time quality metrics via REST APIs.
- Improved trust, reliability, and audit readiness of healthcare pipelines.

EDUCATION

- Master's in Computer Science, Texas Tech University, TX